

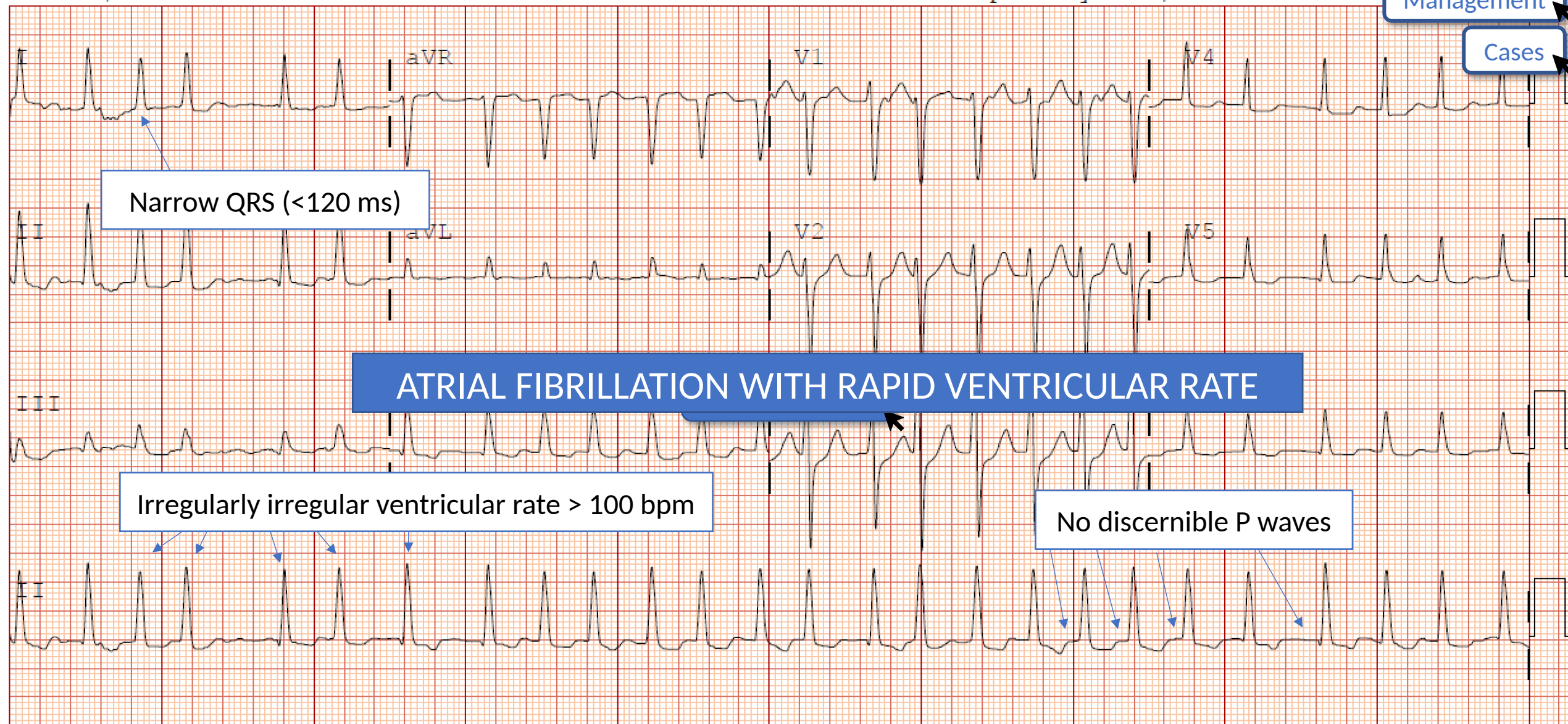
68 yo M with hx of HTN, DMII and COPD who presents with dyspnea on exertion and palpitations.

Pathophysiology

Work-up

Management

Cases



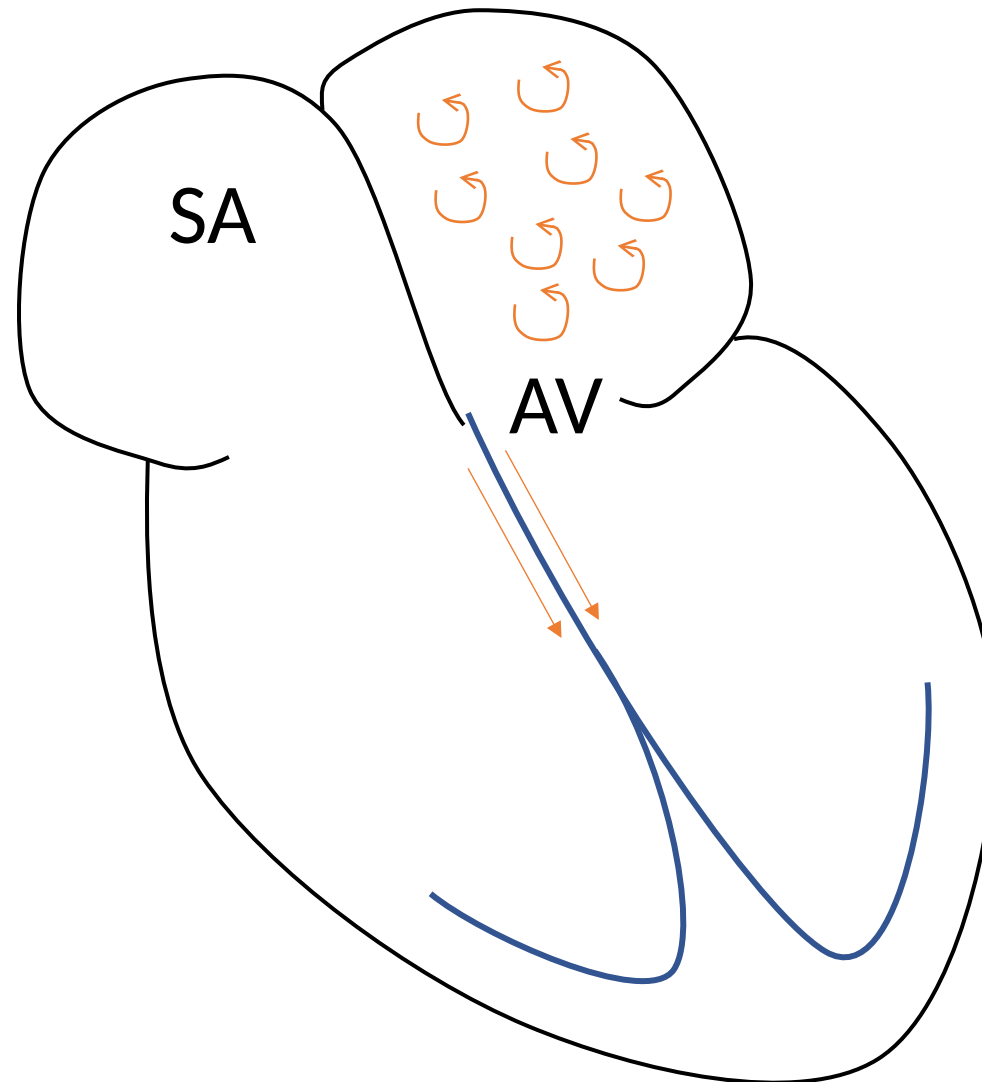
PATHOPHYSIOLOGY

Pathophysiology

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Cases



AF is characterized by disorganized, irregular atrial activity comprised of multiple small reentrant circuits. The atrial rate can exceed > 400 bpm.

Because of the refractory period of the AV node, not all of the atrial impulses are conducted to the ventricles. The ventricular rate in AF determined by the AV node refractory period.

WORK-UP

Pathophysiology

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Cases

Cardiac

Valvular heart disease
Structural heart disease

Decompensated HF

New ischemia

TTE

± BNP

ECG ±
troponin

SA

AV

Non-cardiac

Pulmonary disease

Drugs/ toxins

Metabolic

Endocrinopathies

TTE, ± CXR,
± OSA screen

± urine tox screen, ETOH level

BMP + Mg, CBC

TSH + reflex T4

Non-cardiac causes of AF can include electrolyte imbalances, thyrotoxicosis, infection/sepsis, volume overload, OSA, COPD, and intoxication or withdrawal from drugs/toxins.

MANAGEMENT

Pathophysiology

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Meds table

Cases

1

UNSTABLE?

- Sustained hypotension (SBP < 90)
- SBP ↓ >30 mmHg + CP/AMS/SOB

Yes

2

No

Rate control

HR < 130

HR > 130

ACUTE

CHRONIC

Oral meds

IV meds

HR goal < 110

⚡ Sedation
DCCV 120-200J
Anticoagulation

Stable AF with RVR is managed with rate control. **Acute** HR goal is <130. IV medications should be used to achieve the acute HR goal. Transition to oral medications after HRs < 110. **Chronic** HR goal is to bring HRs < 110s.

SA

AV

BB

CCB

Inotropy

Amio

Digoxin

+

BB

Metoprolol and esmolol block the AV node but are also **negative** inotropes (avoid in decompensated HFrEF).

CCB

Diltiazem blocks the AV node and is a **negative** inotrope. It should be avoided in patients with any heart failure.

Digoxin

Digoxin acts at the Na/K ATPase on cell surfaces. The net effect is that it blocks the AV node and is a **positive** inotrope.

Amiodarone

Amiodarone is a class III antiarrhythmic that acts to block the AV node.

Click for detailed table

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Meds table

Cases

1st line

2nd line

Metoprolol

IV

2.5 - 5 mg IV q5 min
(max 15 mg)

PO

12.5-37.5 mg PO q6h

Diltiazem

Bolus: 10 – 20 mg
Drip rate: 5-15 mg/h

30-90 mg PO q6h

Digoxin

Load: 0.25 – 0.5 mg IV,
*0.25 mg IV q6h x 2**
**renally dosed*

0.5 mg x 1, then 0.25
mg q6h x 1-3 doses

Amiodarone

Load: 150 mg IV push
Drip rate: per protocol

BP effect

↓ ↓

↓ ↓

↔

↓



Caution

Decompensated HFrEF
Sepsis

HFrEF (any history)
Sepsis

Elderly
Renal failure
HypoK/Mag

Chronic AF

Case 1

68 yo M with hx of HTN and DMII presents with new onset AF with RVR with HR of 160s. BPs are 120s/60s. He reports dyspnea and palpitations.

He has stable AF with RVR. HRs > 130, give IV medications (e.g., metoprolol 5 mg or diltiazem 10 mg) to bring down HR. Work-up of RVR and new AF: CBC, BMP/Mg, TSH, ECG, Troponin, CXR

He receives IV metoprolol 5 mg and his HRs improve to 140s. BPs are 110/70s.

HRs remain > 130s with stable BPs. Give another dose IV metoprolol 5 mg.

He receives IV metoprolol 5 mg and his HRs improve to 120s. BPs are 100/70s.

His HRs are now < 130 so he can be started on oral medications - metoprolol 12.5 – 25 mg q6h. His CHADS-VASc is 3, consider anticoagulation.

Case 2

54 yo F with HTN and ETOH use disorder is admitted for septic shock on vasopressors. You are called for new onset AF with HRs in the 150s. MAPs are 70s on 0.1 of norepinephrine.

Despite being on vasopressors, this patient is in stable AF with RVR. Given septic shock, amiodarone is favored over metoprolol or diltiazem.

Case 3

68 yo M with AF with RVR currently on an esmolol drip in the ICU. You are called for HRs in the 160s, BPs that dropped to 85/palp, and the patient is unresponsive.

This patient is unstable! Place pads and prepare for DCCV with 200 J. Sedation/analgesia should be handy but should not delay care. All cardioverted patients should receive anticoagulation.

Case 4

72 yo M with DMII, HTN, HFrEF (25%) presents with AF with RVR with HRs of 160s. BPs 110s/60s. He denies orthopnea and PND.

He has stable AF with RVR. Given HFrEF, he should not receive CCBs. B-blockers can be used in compensated HFrEF. Give IV metoprolol 5 mg.

He receives IV metoprolol 5 mg x 3 and his HRs improve to 140s. BPs are 90/60s. He reports palpitations, but no chest pain, dyspnea, or altered mentation. BMP is normal. Urine output is adequate.

He remains stable but has low BPs from IV metoprolol. Digoxin is the only BP neutral medication and has positive inotropy and should be used as your next agent. Give an IV load of 0.25 – 0.5 mg digoxin.

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